Probabilistic Combinatorics

Sheet 2

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Problem 1. Let $n \geq 2$ and $\mathcal{A}$ a set of n-element subsets of $\mathbb{N}$ with $\#\mathcal{A} < 4^{n-1}$. Show: There is a map $f : \mathbb{N} \to [4]$ with $\#f(A) \geq 2$ for all $A \in \mathcal{A}$.

Proof.
Let $(\Omega, \mathbb{P})$ be the discrete probability space where the sample space Ω is the set of all functions $f : \mathbb{N} \to [4]$ and the probability measure $\mathbb{P}$ is given by independent choices in $[4]$ that are uniformly distributed. Fixed a set $A \in \mathcal{A}$, let

$$\Omega_A = \{f \in \Omega \mid \#f(A) = 1\}$$

This implies that any function $f \in \Omega_A$ must have exactcly one element $i \in [4]$ such that $f(a) = i$ for any $a \in A$. Consider any element $i \in [4]$, the probability of functions $f \in \Omega$ such that $f(a) = i$ for any $a \in A$ is $(\frac{1}{4})^n$. Additionally, there are $\binom{4}{1}$ ways to pick an element $i \in [4]$. Therefore,

$$\begin{aligned} \mathbb{P}(\Omega_A) &= \binom{4}{1}\left(\frac{1}{4}\right)^n \\ &= \frac{1}{4^{n-1}} \end{aligned} \tag{1.1}$$

Furthermore, we have

$$\mathbb{P}(\bigcup_{A \in \mathcal{A}} \Omega_A) \leq \sum_{A \in \mathcal{A}} \mathbb{P}(\Omega_A) \tag{1.2}$$

It follows from $\#\mathcal{A} < 4^{n-1}$ and the equation (1.1) that

$$\begin{aligned} \sum_{A \in \mathcal{A}} \mathbb{P}(\Omega_A) &< 4^{n-1}\frac{1}{4^{n-1}} \\ &< 1 \end{aligned} \tag{1.3}$$

Due to the two inequalities (1.2) and (1.3), we have

$$\mathbb{P}(\bigcup_{A \in \mathcal{A}} \Omega_A) < 1$$

This implies that not every function $f \in \Omega$ has the property where $\#f(A) = 1$ for at least one $A \in \mathcal{A}$. Therefore, there must exist a map $f : \mathbb{N} \to [4]$ such that $\#f(A) \geq 2$ for all $A \in \mathcal{A}$. □

Problem 2. Let $G = (V, E)$ be an undirected graph. Show via a probabilistic argument: There is a subset $E' \subseteq E$ such that $\#E' \geq \#E/2$ and the subgraph $G' := (V, E')$ is bipartite.

Proof.
Let $(\Omega, \mathbb{P})$ be the discrete probability space where the sample space $\Omega = 2^V$ and the probability measure $\mathbb{P}(w) = \frac{1}{2^{|V|}}$ for any $w \in \Sigma$. Next, define the random variable $X : \Omega \to \mathbb{R}$ that maps

any subset of vertices $w \in \Omega$ to number of edges in G that are incident to only one vertex in w. Additionally, for any edge e in G, define the random variable $X_e : \Omega \to \mathbb{R}$ defined by the following rule for any $w \in \Omega$

$$X_e(w) = \begin{cases} 1 \text{ if } e \text{ is incident to only one vertex in } w \\ 0 \text{ otherwise} \end{cases}$$

Now, define the indicator variable $X_e = 1_{A_e}$ of event

$$A_e = \{w \in \Omega \mid e \text{ is incident to only one vertex in } w\}$$

then $\mathbb{E}(X_e) = \mathbb{E}(1_{A_e}) = \mathbb{P}(A_e)$. Let any edge e in G denoted by $e = (u, v)$, then $\mathbb{P}(A_e)$ is calculated as follows

$$\begin{aligned} \mathbb{P}(A_e) &= \mathbb{P}(\{w \in \Omega \mid u \in w \text{ and } v \notin w\}) + \mathbb{P}(\{w \in \Omega \mid u \notin w \text{ and } v \in w\}) \\ &= \left(\frac{1}{2} \times \frac{1}{2}\right) + \left(\frac{1}{2} \times \frac{1}{2}\right) \\ &= \frac{1}{2} \end{aligned}$$

Note that, $X = \sum_{e \in E(G)} X_e$. By linearity of expectation, we have

$$\mathbb{E}(X) = \sum_{e \in E(G)} \mathbb{E}(X_e) = \sum_{e \in E(G)} \mathbb{P}(A_e) = \frac{|E|}{2}$$

Therefore,

$$\mathbb{P}(X > \frac{|E|}{2}) > 0 \text{ or } \mathbb{P}(X = \frac{|E|}{2}) = 1$$

It is worth to note that, for any set of vertices $w \in \Omega$, then any set of edges in G, denoted by $E^{'}$, where each edge is incident to only one vertex in w must form a bipartite graph $G^{'} \coloneqq (V, E^{'})$. That is, due to the inequality and equality shown above, there is a subset $E^{'} \subseteq E$ such that $\#E^{'} \geq \#E/2$ and the subgraph $G^{'} \coloneqq (V, E^{'})$ is bipartite. □

Problem 3. Let $\langle \cdot, \cdot \rangle$ be the standard dot product on $\mathbb{R}^n$. Show probabilistically: For vectors $v_1, \ldots, v_n \in \mathbb{R}^n$ with length $\|v_i\| \coloneqq \sqrt{\langle v_i, v_i \rangle} = 1$ for $1 \leq i \leq n$, there are $\epsilon_1, \ldots, \epsilon_n \in \{\pm 1\}$ with

$$\|\epsilon_1 v_1 + \ldots + \epsilon_n v_n\| \leq \sqrt{n}$$

Proof.
Let $(\Omega, \mathbb{P})$ be the discrete probability space where Ω is the set of all possible choices for $\epsilon_1, \ldots, \epsilon_n$ with the probability measure $\mathbb{P}(w) = \frac{1}{2^n}$ for any $w \in \Omega$. Let $X : \Omega \to \mathbb{R}$ be the random variable that assigns a choice of $\epsilon_1, \ldots, \epsilon_n$ to $\|\epsilon_1 v_1 + \ldots + \epsilon_n v_n\| \leq \sqrt{n}$. That is, we have

$$\begin{aligned} \mathbb{E}(X) &= \mathbb{E}(\|\epsilon_1 v_1 + \ldots + \epsilon_n v_n\| \\ &= \mathbb{E}(\sum_{i=1}^{n} \epsilon_i^2 \|v_i\|^2 + \sum_{1 \leq i < j \leq n} \epsilon_i \epsilon_j \langle v_i, v_j \rangle) \end{aligned}$$

Since $\epsilon_i = \{\pm 1\}$ for $1 \leq i \leq n$, then $\epsilon_i^2 = 1$. The equation then becomes

$$\begin{aligned} \mathbb{E}(X) &= \mathbb{E}(\sum_{i=1}^{n} \|v_i\|^2 + \sum_{1 \leq i < j \leq n} \epsilon_i \epsilon_j \langle v_i, v_j \rangle) \\ &= \sum_{i=1}^{n} \|v_i\|^2 + \mathbb{E}(\sum_{1 \leq i < j \leq n} \epsilon_i \epsilon_j \langle v_i, v_j \rangle) \\ &= \sum_{i=1}^{n} \|v_i\|^2 + \sum_{1 \leq i < j \leq n} \mathbb{E}(\epsilon_i \epsilon_j \langle v_i, v_j \rangle) \end{aligned} \tag{3.1}$$

We can observe from the equation that ϵ_i and ϵ_j are actually two random variables where ϵ_i and ϵ_j are two functions from Ω to $\{\pm 1\}$. Since $\langle v_i, v_j \rangle$ is a constant, the equation (3.1) equivalent to

$$\mathbb{E}(X) = \sum_{i=1}^{n} \|v_i\|^2 + \sum_{1 \leq i < j \leq n} \mathbb{E}(\epsilon_i \epsilon_j) \langle v_i, v_j \rangle \tag{3.2}$$

It is obvious that ϵ_i and ϵ_j are independent from each other. Therefore, $\mathbb{E}(\epsilon_i \epsilon_j) = \mathbb{E}(\epsilon_i)\mathbb{E}(\epsilon_j)$. Consider $\mathbb{E}(\epsilon_i)$, we have

$$\mathbb{E}(\epsilon_i) = \sum_{w \in \Omega} \epsilon_i(w) \mathbb{P}(w)$$

Recall that the image of the random variable ϵ_i is the set $\{\pm 1\}$. Therefore, the equation is actually equivalent to

$$\begin{aligned}
\mathbb{E}(\epsilon_i) &= \sum_{w \in \Omega | \epsilon_i(w) = 1} \mathbb{P}(w) - \sum_{w \in \Omega | \epsilon_i(w) = -1} \mathbb{P}(w) \\
&= \mathbb{P}(\{w \in \Omega | \epsilon_i(w) = 1\}) - \mathbb{P}(\{w \in \Omega | \epsilon_i(w) = -1\}) \\
&= \frac{1}{2} - \frac{1}{2} \\
&= 0
\end{aligned}$$

Similarly, we have $\mathbb{E}(\epsilon_j) = 0$, and thus $\mathbb{E}(\epsilon_i \epsilon_j) = 0$. Therefore, the equation (3.2) is equivalent to

$$\mathbb{E}(X) = \sum_{i=1}^{n} \|v_i\|^2$$

Recall that $\|v_i\|^2 = 1$ for $1 \leq i \leq n$, thus $\mathbb{E}(X) = n$. This implies that $P(X < n) > 0$, which means that there exists a choice of $\epsilon_1, \ldots, \epsilon_n \in \{\pm 1\}$ such that

$$\|\epsilon_1 v_1 + \ldots + \epsilon_n v_n\|^2 \leq n \Leftrightarrow \|\epsilon_1 v_1 + \ldots + \epsilon_n v_n\| \leq \sqrt{n}$$

and the proof is finished. □

Problem 4. Let $(\Omega, \mathbb{P})$ be a discrete probability space and let $T, X_1, X_2 \ldots$ be real valued random variables on $(\Omega, \mathbb{P})$ such that:

- All expected values exist
- The random variables are independent
- The random variables $X_1, X_2, \ldots$ are identically distributed
- $T(\omega) \in \mathbb{N}$ for all $\omega \in \Omega$

Prove that:

(i) Let $T_n, n \in \mathbb{N}$ be the indicator random variable of the event $\{\omega \in \Omega | T(\omega) = n\}$ and $S_n := X_1 + X_2 + \ldots + X_n$. Then T_n and S_n are independent

(ii) Define the random variable

$$\begin{aligned}
S_T : \Omega &\to \mathbb{R} \\
: \omega &\mapsto \sum_{i=1}^{T(w)} X_i(w).
\end{aligned}$$

Show that $S_T = \sum_{n \in \mathbb{N}} S_n \cdot T_n$

(iii) $\mathbb{E}[S_T]$ exists and $\mathbb{E}[S_T] = \mathbb{E}[T]\mathbb{E}[X_1]$

Proof.
We are going to prove the statements one after another

(i) In order to prove that T_n and S_n are independent, we want to show that

$$\mathbb{P}(T_n = a, S_n = b) = \mathbb{P}(T_n = a)\mathbb{P}(S_n = b) \tag{4.1}$$

We firstly consider the case where $T_n = 1$, we have

$$\begin{aligned}
\mathbb{P}(T_n = 1, S_n = b) &= \mathbb{P}(\{w \in \Omega | T(w) = n \text{ and } S_n(w) = b\}) \\
&= \mathbb{P}(\{w \in \Omega | T(w) = n\} \cap \{w \in \Omega | S_n(w) = b\}) \\
&= \mathbb{P}(\{w \in \Omega | T(w) = n\} \cap (\bigcup_{x_1+x_2+\ldots+x_n=b} \bigcap_{i=1}^{n} \{w \in \Omega | X_i = x_i\})
\end{aligned}$$

By the distributive property of intersection over union, we have

$$\begin{aligned}
\mathbb{P}(T_n = 1, S_n = b) &= \mathbb{P}(\bigcup_{x_1+x_2+\ldots+x_n=b} (\{w \in \Omega | T(w) = n\} \cap \bigcap_{i=1}^{n} \{w \in \Omega | X_i = x_i\})) \\
&= \sum_{x_1+x_2+\ldots+x_n=b} \mathbb{P}(\{w \in \Omega | T(w) = n\} \cap \bigcap_{i=1}^{n} \{w \in \Omega | X_i = x_i\})
\end{aligned}$$

Since $T, X_1, X_2, \ldots$ are independent, we have

$$\begin{aligned}
\mathbb{P}(T_n = 1, S_n = b) &= \sum_{x_1+x_2+\ldots+x_n=b} \mathbb{P}(\{w \in \Omega | T(w) = n\}) \prod_{i=1}^{n} \mathbb{P}(\{w \in \Omega | X_i = x_i\}) \\
&= \mathbb{P}(\{w \in \Omega | T(w) = n\}) \sum_{x_1+x_2+\ldots+x_n=b} \prod_{i=1}^{n} \mathbb{P}(\{w \in \Omega | X_i = x_i\}) \\
&= \mathbb{P}(\{w \in \Omega | T(w) = n\}) \mathbb{P}(\{w \in \Omega | S_n(w) = b\}) \\
&= \mathbb{P}(T_n = 1)\mathbb{P}(S_n = b)
\end{aligned}$$

The same thing can be shown for $T_n = 0$ in the similar way. Therefore, the equation (4.1) is satisfied, and thus T_n and S_n are independent.

(ii) For any $\omega \in \Omega$, let $n = T(w)$. Then $S_T(w) = \sum_{i=1}^{n} X_i(w) = S_n(w)$. Furthermore, for any $w \in \Omega$, then $T(w)$ is unknown. That is, for any $w \in \Omega$, we want to try out all $S_n(w)$ for $n \in \mathbb{N}$ and only take the one where $T(w) = n$, or $T_n(w) = 1$. Additionally, $T_n(w) = 0$ when $T(w) \neq n$, and thus $S_T(w)$ can be expressed as $S_T(w) = \sum_{n\in\mathbb{N}} S_n(w)T_n(w)$ for any $w \in \Omega$. Therefore,

$$S_T = \sum_{n\in\mathbb{N}} S_n T_n$$

(iii) Basaed on the statement (ii) shown above, we have

$$\begin{aligned}
\mathbb{E}(S_T) &= \mathbb{E}(\sum_{n\in\mathbb{N}} S_n T_n) \\
&= \sum_{n\in\mathbb{N}} \mathbb{E}(S_n T_n)
\end{aligned}$$

Since S_n and T_n are independent according to the statement (i), the equation is equivalent to

$$\begin{aligned}
\mathbb{E}(S_T) &= \sum_{n\in\mathbb{N}} \mathbb{E}(S_n)\mathbb{E}(T_n) \\
&= \sum_{n\in\mathbb{N}} \mathbb{E}(X_1 + X_2 + \ldots + X_n)\mathbb{E}(T_n) \\
&= \sum_{n\in\mathbb{N}} [\mathbb{E}(X_1) + \mathbb{E}(X_2) + \ldots + \mathbb{E}(X_n)]\mathbb{E}(T_n)
\end{aligned}$$

Since the random variables $X_1, X_2, \ldots, X_n$ on the probability space $(\Omega, \mathbb{P})$ are identically distributed. The equation is equivalent to

$$\begin{aligned}\mathbb{E}(S_T) &= \sum_{n\in\mathbb{N}} n\mathbb{E}(X_1)\mathbb{E}(T_n) \\ &= \mathbb{E}(X_1)\sum_{n\in\mathbb{N}} n\mathbb{P}(T=n) \\ &= \mathbb{E}(X_1)\mathbb{E}(T)\end{aligned}$$

and the proofs for the statements are finished. □